

1 Superspace and Information Projection

Define a superspace $\mathcal{S}$ representing the full, “raw” information content of the universe:

$$\mathcal{S} = \{I_i\}_{i \in N}, \quad I_i = \text{fundamental information modes}$$

Introduce a projection *operator* $\Pi: \mathcal{S} \rightarrow \mathcal{M}$, where $\mathcal{M}$ is the emergent 3D metric space (observable universe):

$$\Pi(I) = \text{mode in physical 3D space}$$

Key properties of Π for stability:

1. Continuity: small perturbations in $\mathcal{S}$ yield small changes in $\mathcal{M}$.
2. Critical bandwidth: $\dim_{\text{spectral}}(\Pi(\mathcal{S})) = D_s^* \approx 2.97$.
3. Partial non-fidelity: Π may lose information ($S_{\text{loss}} > 0$), generating attractors.

2 Spectral Structure and Laplacian

On $\mathcal{M}$, define a Laplacian operator Δ governing spectral modes:

$$\Delta \phi_n = \lambda_n \phi_n, \quad \{\phi_n\} \text{ orthonormal basis}$$

- λ_1 = first nonzero eigenvalue, sets scale of attractors.
- Higher λ_n correspond to finer structures.

The dimension spectral function:

$$D_s(k) = -2 \frac{d \log \text{Tr} [e^{-k\Delta}]}{d \log k}$$

- Flow of $D_s(k)$ along scale k encodes the RG trajectory.
- Plateau $D_s \approx 2.97$ generates the stable attractor band.

3 Flot Fractal / RG Flow

Define a gradient-like flow on the space of metrics $\mathcal{M}$, spectral modes, and internal algebra:

$$\frac{dg(t)}{dt} = -\nabla \mathcal{F}[g(t), \{\lambda_n\}, \mathcal{A}], \quad g \in \text{metrics}, \quad \mathcal{A} \text{ internal algebra}$$

- $\mathcal{F}$ = global information loss functional (entropy + spectral constraints).
- This flow satisfies an Evolution Variational Inequality (EVI):

$$\frac{1}{2} \frac{d}{dt} |g(t) - h|^2 + \mathcal{F}[g(t)] \leq \mathcal{F}[h], \quad \forall h$$

- Guarantees convexity, uniqueness, and stability of attractors.

4 Correlations

Functional decomposition of information loss:

$$\mathcal{F} = \sum_i f_{\text{direct}}(\Pi(I_i), \Pi(I_j)) + \sum_{i,j,k} f_{\text{indirect}}(\Pi(I_i), \Pi(I_j), \Pi(I_k)) \\ + f_{\text{topo}}(\{\Pi(I_i)\})$$

1. Direct: local pairwise attraction (electron $\leftrightarrow$ proton).
2. Indirect: multi-mode interactions, stabilize molecules.
3. Topological: global constraints (CD(K,D*), EVI convexity, D_s plateau).

5 Algebraic Internal Structure

Define internal algebra $\mathcal{A}$ acting on spectral subspaces:

$$\mathcal{A} = M_3(C) \quad (\text{fermion family structure})$$

- Representation splits chiral subspaces, generating left-right asymmetry.
- Global EVI + flow constraints ensure minimal, unique algebra compatible with observed 3 families.

6 Couplings and Constants

Spectral closure fixes physical constants:

$$\Lambda \sim f(\lambda_1, D^*$$

$$s),$$

$$G \sim h(\Lambda, \text{saturation constraints}),$$

$$g_1, g_2, g_3 \sim \text{unification from spectral ratios},$$

$$m_f \sim \text{critical eigenvalues of Laplacian projected onto fermionic subspaces}$$

- The constants emerge from spectral saturation + entropic convexity.
- Higgs criticality is encoded in topological and spectral constraints.

7 Dynamics of Particles and Molecules

Attractors for electron, atom, and molecule:

$$\text{Attractor} = \arg \min_{\Pi(I_i)} \mathcal{F} [\Pi(I_i), g, \mathcal{A}]$$

- Minimizes information loss, reproducing bound states naturally.
- Multi-particle systems arise via indirect and topological correlations.

8 Emergent Time

- Time t is emergent, parameterizing the evolution along the gradient flow.
- Relates information processing to physical dynamics:

$$\frac{d}{dt} \Pi(I) = -\nabla_{\Pi(I)} \mathcal{F}$$

- *Past/future* = *positions* along flow; present = instantaneous attractor configuration.

9 Global Properties

1. Bande critique: attractors form a stable band for $D_s^* \in [2.7, 2.97]$.
2. Stability cosmological / entropic: convexity + $CD(K, D^*)$ ensure no Big Rip, bounded total entropy.
3. Universality: unique (or discrete) solutions for internal algebra, constants, and spectra.

Summary

$$\boxed{\text{MPFS Model} \equiv (\mathcal{S}, \Pi, \Delta, \mathcal{A}, g, \mathcal{F}, D_s^*, \text{flow } t)}$$

- Superspace $\mathcal{S} \rightarrow \text{projection } \Pi \rightarrow 3\text{D spectral modes}$.
- Gradient flow minimizes information loss functional $\mathcal{F}$ (EVI + correlations).
- Spectral and algebraic constraints fix constants, attractors, and particle structure.

- Emergent time and hierarchy of attractors reproduce electron, atom, molecule, and cosmological structures.